

Contextual Structure-Enhanced Selective Graph Convolutional Network

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Abstract

Graph Neural Networks fundamentally rely on homophily assumptions where connected nodes are expected to share similar labels, consequently suffering severe performance degradation in heterophilic graphs due to the indiscriminate neighbor aggregation mechanism. Although recent solutions have attempted to incorporate higher-order neighborhoods or reweighting schemes, they often inadvertently amplify structural noise by introducing a larger proportion of dissimilar nodes than similar ones, while simultaneously failing to capture nuanced contextual patterns due to their inability to discern subtle local structural variations across subgraphs. To holistically address these intractable and co-existing challenges, we propose the Contextual Structure Enhanced Selective Graph Convolutional Network (CSS-GCN), a novel architecture that organically synergizes contextual structure modeling with adaptive neighbor selection. Specifically, our approach employs ego-network partitioning and group fairness constraints to effectively quantify

domain-invariant structural patterns, thereby countering the contextual blindness often observed in conventional GNNs. Complementarily, we design a selective propagation mechanism unifying adaptive neighborhood distribution-based similarity computation with the gated fusion of three distinct information pathways: potential homophilic neighbors identified through attribute-topology synergy, first-hop connections, and ego-representations. This dual-component framework enables nodes to dynamically filter out irrelevant signals while preserving structural consistency across diverse homophily-heterophily landscapes. Extensive validation on 10 real-world graphs demonstrates the effectiveness and superiority of our proposed approach.

CCS Concepts

• Computing methodologies → Machine learning algorithms.

Keywords

Graph neural network, Node classification, Information transmission, Graph convolutional neural network

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1 Introduction

Graph structures, which are ubiquitous across a vast array of diverse real-world scenarios, stand as exceptionally powerful and versatile tools for modeling intricate and complex relational data [30]. As a dominant and widely adopted methodology within the broader field of graph representation learning, graph neural networks (GNNs) have been the subject of extensive and in-depth investigation. In recent years, GNNs have garnered substantial and growing attention, largely due to their remarkable and robust capabilities in effectively addressing a wide range of diverse real-world applications that involve graph-structured data; these applications include, but are not limited to, node classification [17], graph classification [9], and link prediction [42]. By seamlessly integrating convolutional architectures with principles from graph signal processing, a multitude of graph convolutional network (GCN) frameworks have been developed, and these frameworks have consistently demonstrated superior and state-of-the-art performance across the aforementioned domains.

Recent studies have suggested that the success of GCNs and their various variants relies fundamentally and critically on the homophily assumption [3], a principle wherein connected nodes share highly identical class labels [11]. This principle provides an essential inductive bias that significantly enhances the generalized message-passing mechanism, a mechanism that aggregates information from neighboring nodes to update node representations—a process widely recognized as a specialized form of low-pass filtering [4]. Most mainstream variants of GNNs, such as GCN, GAT [33], JKNet [37], PCNet [18], and GPRGNN [7], adopt message-passing architectures to learn compact and effective node embeddings through the iterative aggregation of neighborhood information.

However, the homophily assumption does not always hold true universally in all scenarios [23]. In such circumstances, a substantial and growing body of empirical evidence has clearly demonstrated that GCN may even perform significantly worse than simple Multi-Layer Perceptrons (MLPs) [25] that rely solely on node features as their input, a phenomenon widely recognized as the heterophily issue. A potential explanation for this unfavorable outcome is that the low-pass filter, which is inherently part of the GCN framework, is believed to exert a profoundly detrimental impact on the model's performance when applied to heterophilic graphs. In fact, a closer examination of real-world graphs reveals that they often inherently exhibit significant heterophilic characteristics, where nodes that are connected to each other tend to belong to different classes. As a direct result of this inherent property, GCN and several of its well-known variants, such as SGC [35] and GAT, may produce classification results that are far from satisfactory when dealing with heterophilic networks. The underlying reason for this subpar performance lies in the fact that message propagation and aggregation processes occurring between nodes belonging to different classes expose these nodes to an excessive amount of irrelevant information, thereby severely undermining the accuracy of their classifications.

Existing methods have extensively sought to adapt to complex heterophilic graphs by strategically incorporating representations of multi-hop neighbors to include a greater number of nodes belonging to the same class. For instance, certain spectral methods,

such as APPNP [10], GCNII [6], incorporate multiple stacked layers, a deep design that effectively expands a node's receptive neighborhood by systematically encompassing multi-hop neighbors within its reach. Furthermore, these methods employ sophisticated initial connection mechanisms, skip connection strategies, or linearization techniques to integrate distinct representations derived from different network layers. In addition, a number of spatial methods also significantly enrich the information from neighboring nodes by iteratively aggregating representations across multi-hop neighbors; examples include MixHop [1], H₂GCN [46], and HopGNN [5], which fully leverage comprehensive information from both first-order and higher-order neighbors to enhance the learning process. Meanwhile, there exist other methods that aim to improve performance from the broader graph-level perspective. Recent research efforts, for instance, have focused on enhancing the applicability of GNNs in challenging transfer learning scenarios. Specifically, DANE [43] employs adversarial training strategies for domain classifiers based on node representations extracted from the last layer, while UDAGCN [36] further introduces inter-graph attention mechanisms on top of such adversarial training frameworks to further strengthen performance. However, these methods often fail to adequately account for the inevitable noise introduced by higher-order neighbors, do not model different node attributes in a sufficiently targeted manner, and underestimate the inherent complexity and diversity of real-world data.

Although homophily varies across different networks, there also exist substantial and often overlooked differences in the internal structural patterns within individual networks [19, 36]. Against this backdrop, we propose the Contextual Structure Enhanced Selective Graph Convolutional Network (CSS-GCN). Specifically, we first employ a meticulously designed self-network construction module for each node within the graph to create subgraphs, where these subgraphs capture a diverse range of local structural information that is essential for the effective transfer of structural knowledge. Building upon this foundation, we then proceed to obtain the high-order neighborhoods and potential neighbors of the graph through the precise calculation of node neighborhood similarity. Next, with the aim of modeling the attributes of different nodes, we have designed an effective multi-channel selective fusion mechanism, which serves to efficiently integrate information from multiple dimensions. And finally, we dynamically balance the divided subgraphs through the implementation of a component loss mechanism. We have conducted extensive experiments on 10 real-world datasets with distinct attributes, thereby verifying the effectiveness and superiority of CSS-GCN. To summarize, the key contributions of this work are as follows:

- (1) Our research enhances the expressive capability of information by means of partitioning graphs into subgraphs, ensuring that sufficient information from neighbors can be fully captured even under sufficiently complex circumstances.
- (2) The multi-channel selective fusion mechanism, along with the component balancing loss, is capable of modeling both homophily and heterophily, allowing each node to selectively receive relevant neighbor information and thus achieve more favorable node representations.

(3) Extensive experiments are carried out on 10 datasets exhibiting different properties, highlighting the superior performance of CSS-GCN.

2 Related Work

2.1 Graph Neural Networks

The development of graph neural networks can be systematically traced back to pioneering early efforts where Defferrard et al. [8] proposed an efficient model framework by extending the principles of convolutional neural networks from regular Euclidean grid structures, such as image data, to irregular non-Euclidean structures, including graphs. Following this foundational work, the Graph Convolutional Network emerged as a widely popular and dominant graph neural network paradigm; it acquires robust node embeddings by recursively integrating high-order neighborhood information through the strategic use of stacked graph convolutional layers. Advancing the methodology further, GraphSAGE [11] significantly expanded the scope of aggregation beyond elementary averaging operations and explicitly modeled the ego-node features separately from the features derived from neighboring nodes within its sub-sampled neighborhood. Additionally, Graph Attention Networks (GAT) [33] introduced sophisticated node-level multi-head attention mechanisms to adaptively assign specific importance weights to information originating from different neighborhoods, thereby enhancing the model's capability to focus on the most relevant nodes.

As the field has matured, subsequent research efforts have been consistently directed toward developing significantly deeper architectures [21, 38] and more scalable frameworks [13, 20] to effectively address various challenges in graph learning. Moreover, building extensively upon the foundations laid by GCN, attention-based methods [14, 32] have been designed to dynamically learn weights, frequently referred to as attention scores, on edges during the message-passing process, effectively incorporating adaptive attention mechanisms into the neighborhood aggregation procedure. These algorithmic advancements have enabled graph neural networks to demonstrate strong performance across a wide array of real-world tasks involving graph-structured data, despite the inherent limitations in capturing long-range dependencies due to the constraints of their finite aggregation layers.

2.2 Homophily and Heterophily

Most existing Graph Neural Networks (GNNs) predominantly operate under the fundamental homophily assumption, which posits that nodes belonging to the same class are significantly more likely to be connected to one another within a graph structure. However, heterophily is a complex and nuanced property that is ubiquitous across a wide range of diverse real-world graph scenarios, where nodes possessing different labels tend to be linked together. This phenomenon serves as a notable constraint that can significantly limit the predictive performance of GNNs specifically designed to work under standard homophilic conditions, often leading to the indistinguishable mixing of node features. In direct response to this challenge, a growing body of recent research has strategically focused on enhancing the capability of GNNs to better handle heterophilic graphs. Among these efforts, some approaches seek to

extend the neighborhood scope of central nodes by systematically incorporating informative high-order neighbors into the aggregation process [15, 46], thereby capturing broader contextual information. Meanwhile, others question the validity of predefined edges and instead aim to discover potential neighboring nodes based on the similarity of their representations [22, 41]. Additionally, a number of new aggregation functions have been specifically designed to more effectively discriminate between node representations in the context of heterophilic graphs, aiming to mitigate the interference of noisy signals [4, 40].

Beyond these initial efforts, several dedicated research initiatives have been directed toward alleviating the limitations imposed by the homophily assumption, with the ultimate goal of improving the robustness and performance of graph convolutional networks. For instance, Geom-GCN [27] introduces a novel geometric aggregation scheme aimed at overcoming the inevitable loss of neighborhood structural information and the potential lack of long-range dependencies in graph data. GPR-GNN [7] proposes a new architectural framework that automatically and adaptively learns generalized PageRank weights, enabling the joint optimization of node feature extraction and topological information capture across different propagation steps. CPGNN [45] designs a graph neural network framework that incorporates an interpretable compatibility matrix H to systematically model the level of homophily or heterophily present in the graph. Furthermore, H_2 GCN [46] seamlessly integrates three key design elements: the separation of ego-node and neighbor embeddings, the inclusion of higher-order neighborhoods, and the combination of intermediate representations. These components work synergistically to capture valuable information even under conditions of low homophily. More recently, GGCN [39] has put forward a robust and generalized model that effectively addresses discrepancies in both features and degrees between neighboring nodes by incorporating signed messages and learned degree corrections, thereby alleviating the reliance of graph convolutional networks on the homophily assumption.

In recent years, a number of sophisticated methods have attempted to derive a refined graph structure from a given original graph, motivated by the observation that raw topology may contain noise. For example, FGC [16] and GloGNN [19] operate under the assumption that the learned graph structure should maintain a close and semantic resemblance to the structural information inherent in the original graph, which is achieved by defining and leveraging higher-order relationships. MCGC [26] employs a graph-level contrastive regularizer to enhance the discriminability of the graph; instead of relying solely on raw topology, it constructs reliable structural views based on highly relevant neighbors for contrastive learning. Drawing inspiration from the successes achieved by such GNNs, a multitude of models have turned their attention to the issue of graph quality when applying these networks [31]; these models either compute edge weights based on node representations or jointly optimize the adjacency matrix alongside the parameters of the GNNs. However, it is worth noting that these methods predominantly focus on homophilic graphs, largely overlooking the critical significance of heterophily in real-world scenarios. This oversight can unfortunately lead to suboptimal performance when such methods are applied to heterophilic graphs, where the structural assumptions may not hold.

Furthermore, some innovative methods strive to adaptively learn a homophilic graph structure from a heterophilic one to facilitate effective message passing. For instance, HOG-GCN [34] is designed to learn a homophilic graph where edge weights are determined by the extent to which nodes belong to the same class across the entire heterophilic graph. In the process of neighbor aggregation, intra-class nodes that exhibit lower levels of homophily are able to exert a proportionally greater influence compared to underlying inter-class nodes. This is achieved by incorporating class-aware information throughout the propagation process. That said, these methods typically rely heavily on node labels to select homophilic edges or guide the structure learning process. This dependence can severely limit their practical applicability in scenarios where labels are scarce or unavailable, such as in semi-supervised or unsupervised learning settings.

3 Method

3.1 Overall Framework

This section provides a detailed exposition of the proposed CSS-GCN framework, whose overall architecture is illustrated in Figure 1. To begin with, the framework constructs subgraphs for each node and identifies potential neighboring nodes. Following this, CSS-GCN extracts a variety of neighbor representations by executing message passing processes across multiple channels. In the final stage, the framework fuses information from multiple dimensions through a multi-channel attention mechanism and the Partition Balanced Loss.

3.2 Preliminary

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected attributed network, where $\mathcal{V}$ is the node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set as $X \in \mathbb{R}^{n \times d}$. Denote by $A \in \mathbb{R}^{n \times n}$ to be the adjacency matrix, and the diagonal matrix standing for the degree matrix such that $D_{ii} = \sum_{j=1}^n A_{ij}$. Let $\tilde{A}$ and $\tilde{D}$ represent matrices that include self-loops. Specifically, $\tilde{A}$ is obtained by adding the identity matrix I to matrix A , and $\tilde{D}$ is obtained by adding the identity matrix I to matrix D .

Pei et al. [27] proposed a metric to quantify the proportion of direct neighbors belonging to the same class as the central node, specifically defining the homophily of a node as the ratio of neighbors with the same label among all its neighbors:

$$\mathcal{H}_i = \frac{|\{v_j \mid (v_i, v_j) \in \mathcal{E}, y_i = y_j\}|}{|d_i|} \quad (1)$$

where y_i is the one-hot encoded representation of the label of node v_i , and d_i denotes the degree of node v_i . The graph-level homophily $\mathcal{H}_G$ is defined as the average value of individual node-level homophily metrics across all nodes within the set $\mathcal{V}$:

$$\mathcal{H}_G = \frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} \mathcal{H}_i \quad (2)$$

The values of both $\mathcal{H}_i$ and $\mathcal{H}_G$ precisely fall within the interval $[0, 1]$. Specifically, when their values are greater than or equal to 0.5, the corresponding node or graph is considered to be homophilic. Conversely, when their values are less than or equal to 0.5, the corresponding node or graph is regarded as heterophilic.

However, the existence of heterophily poses a significant challenge for standard GNNs, which typically rely on the assumption that connected nodes share similar labels. In scenarios with low homophily, indiscriminately aggregating information from direct neighbors introduces noise and degrades performance. To address this limitation, we argue that a robust model must transcend the limitations of immediate connectivity by explicitly modeling the localized contextual structure of each node. Motivated by this, we propose to construct k -hop ego-networks and utilize a selective aggregation mechanism that fuses topological and semantic cues. This strategy allows the model to dynamically adapt to diverse structural patterns, providing a solid foundation for the methodology detailed in the subsequent sections.

3.3 Neighbors Aggregation

Building upon the graph partitioning strategy outlined in Section 3.1, our framework is intimately associated with the constructed k -hop ego-networks. Recent advances in graph learning [2, 44] have highlighted that prevalent message-passing mechanisms are susceptible to noise when the homophily assumption is violated. In scenarios with low homophily, indiscriminately aggregating information from direct topological neighbors often leads to the propagation of misleading signals and feature over-smoothing. Inspired by these insights, we employ a selective aggregation mechanism designed to filter out heterophilic noise by synthesizing topological and attribute evidence. Within the local scope of each ego-network, we first establish a robust metric to identify truly informative neighbors. We define the Structural Relevance, denoted as $\mathcal{R}_{struc}(u, v)$, to quantify the connectivity tightness between the central node u and a candidate neighbor v . This metric relies on the theoretical premise that nodes sharing a dense set of common neighbors are statistically more likely to belong to the same latent community, providing a topological signal that is robust against sporadic heterophilic connections:

$$\mathcal{R}_{struc}(u, v) = |\mathcal{N}(u) \cap \mathcal{N}(v)| \quad (3)$$

where $\mathcal{N}(u)$ represents the immediate neighbor set of node u . While structural overlap provides strong topological evidence, it may be insufficient in sparse regions where structural signals are weak. To mitigate this limitation, we complement the structural view with the Attribute Relevance metric, $\mathcal{R}_{attr}(u, v)$, which evaluates the semantic consistency between the aggregated neighborhood features:

$$\mathcal{R}_{attr}(u, v) = \text{Cos} \left(\sum_{i \in \mathcal{N}(u), i \neq u} \mathbf{x}_i, \sum_{j \in \mathcal{N}(v), j \neq v} \mathbf{x}_j \right) \quad (4)$$

where $\mathbf{x}_i$ denotes the attribute vector of node i . By calculating the cosine affinity of aggregated features, we capture the semantic alignment that often persists even when structural connections are noisy. These two perspectives are fused into a comprehensive Similarity Matrix S :

$$S(u, v) = \lambda \mathcal{R}_{struc}(u, v) + (1 - \lambda) \mathcal{R}_{attr}(u, v) \quad (5)$$

Here, $\lambda \in [0, 1]$ is a tunable hyperparameter acting as a homophily-aware filter. Theoretically, this parameter balances the reliance on structural priors versus feature smoothness. By adjusting λ , the model can adaptively shift its focus from topological evidence in

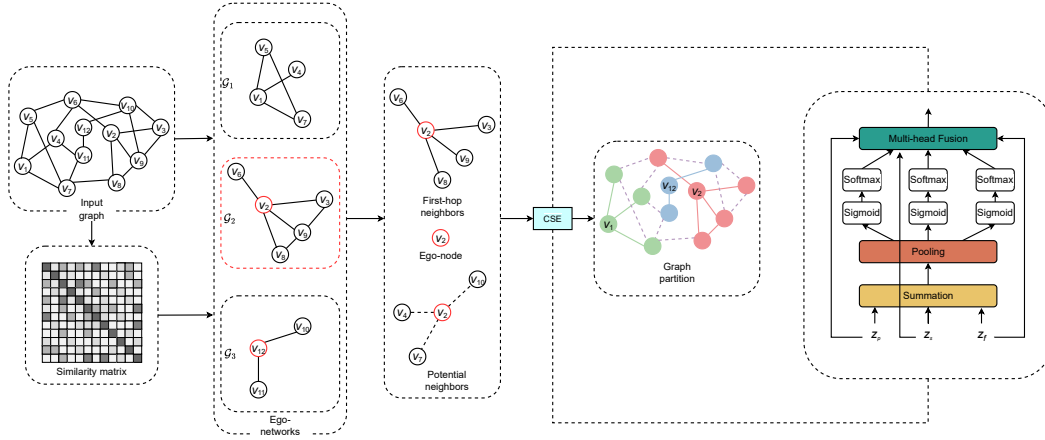


Figure 1: The framework of CSS-GCN.

homophilic regions to semantic similarity in heterophilic boundaries. To ensure computational efficiency and strictly bound the propagation of noise, we preserve only the top- k edges with the highest scores in S , thereby constructing a sparse and high-quality potential neighbor structure.

Based on the refined local structure, we seek to learn discriminative node representations. We decouple the information propagation into three parallel canonical pathways to capture distinct structural properties and prevent the loss of idiosyncratic information. The Potential Context Pathway aggregates information from the discovered high-relevance nodes to capture long-range semantic dependencies:

$$z_p^{(l)} = \sigma(\hat{D}_p^{-1/2} A_p \hat{D}_p^{-1/2} z^{(l-1)} W_p^{(l)}) \quad (6)$$

where A_p is the adjacency matrix derived from the top- k similarity selection and $\hat{D}_p$ is its degree matrix. Simultaneously, the Local Connectivity Pathway preserves the original physical topology to maintain structural integrity and local proximity:

$$z_f^{(l)} = \sigma(\hat{D}^{-1/2} A \hat{D}^{-1/2} z^{(l-1)} W_f^{(l)}) \quad (7)$$

where A represents the original immediate connections. Finally, the Ego-Preservation Pathway retains the intrinsic features of the node to prevent information washout during deep propagation:

$$z_s^{(l)} = \sigma(z^{(l-1)} W_s^{(l)}) \quad (8)$$

where $z^{(0)} = \mathbf{x}$. This multi-channel design ensures that the model comprehensively integrates potential homophily, local topology, and self-information, providing a robust and theoretically grounded basis for the subsequent adaptive fusion module.

3.4 Contextual Structure Enhanced

To effectively synthesize the heterogeneous information captured within the constructed k -hop ego-networks, we employ a dynamic fusion mechanism. Unlike static aggregation, this module acts as a consensus system, evaluating the importance of potential (z_p), first-order (z_f), and self (z_s) representations relative to the holistic subgraph context.

Formally, we first consolidate the representations from the three pathways into a unified context vector to abstract the global state of the ego-network. For the h -th attention head, a global query vector $\mathbf{q}_h^{(l)}$ is derived as follows:

$$\begin{aligned} \mathcal{U}^{(l)} &= z_p^{(l)} + z_f^{(l)} + z_s^{(l)}, \\ \mathbf{q}_h^{(l)} &= \frac{1}{N} \sum_{i=1}^N (\mathcal{U}^{(l)} W_{q,h}^{(l)}) \end{aligned} \quad (9)$$

where $W_{q,h}^{(l)}$ projects the unified features into the query space. To determine the channel-specific relevance, we perform a projection of each node's view-specific representation against this global query. The unnormalized relevance score $\eta_{*,h}^{(l)}$ for each view $*$ $\in \{p, f, s\}$ is computed via sigmoidal gating:

$$\eta_{*,h}^{(l)} = \sigma(z_*^{(l)} (\mathbf{q}_h^{(l)})^\top) \quad (10)$$

where $\sigma(\cdot)$ denotes the sigmoid activation. These scores are then refined through a channel-mixing transformation and normalized to produce the attention distribution $\pi_h^{(l)}$:

$$\pi_h^{(l)} = \text{softmax} \left(\frac{[\eta_{p,h}^{(l)}, \eta_{f,h}^{(l)}, \eta_{s,h}^{(l)}] W_{v,h}^{(l)}}{\tau} \right) \quad (11)$$

where τ scales the logits to control the sharpness of the attention distribution, and $W_{v,h}^{(l)}$ allows for interaction between channels.

To mitigate the bias of single-head estimation, we aggregate the independent assessments from all H heads. The final consensus weights $\bar{\alpha}_*^{(l)}$ are obtained by averaging the head-specific distributions:

$$\bar{\alpha}_*^{(l)} = \frac{1}{H} \sum_{h=1}^H \pi_{*,h}^{(l)}, \quad \forall * \in \{p, f, s\} \quad (12)$$

Finally, the node representation $z^{(l)}$ is updated by dynamically integrating the information sources based on these consensus weights:

$$z^{(l)} = \sum_{\kappa \in \{p, f, s\}} \bar{\alpha}_\kappa^{(l)} z_\kappa^{(l)} \quad (13)$$

This summation formulation ensures that the model adaptively prioritizes the most discriminative structural context for each node, leveraging the diversity preserved in the ego-networks.

3.5 Partition Balanced Loss

To execute the node classification task, the learned representations $z^{(L)}$ are passed through a softmax layer to generate the predictive probability distribution over the class set, formulated as $\hat{Y} = \text{softmax}(z^{(L)})$. The optimization objective focuses on minimizing the cross-entropy loss calculated over the labeled nodes $\mathcal{V}_L$:

$$\mathcal{L}_{CE} = - \sum_{v_i \in \mathcal{V}_L} \sum_{j=1}^C Y_{ij} \ln \hat{Y}_{ij} \quad (14)$$

where C represents the total number of classes. However, relying solely on standard global optimization often leads to structural overfitting, a phenomenon where the model disproportionately prioritizes dominant patterns prevalent in the graph while neglecting underrepresented or complex topologies. This tendency results in poor generalization on nodes residing in structurally distinct or minority regions. To rectify this optimization imbalance and ensure robust generalization across diverse local contexts, we employ a graph partitioning strategy that divides the nodes into M non-overlapping subgraphs: $\{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_M\}$.

For each subgraph $\mathcal{G}_m$, we impose a Structure-Aware Balanced Objective to enforce group-level fairness during the training process:

$$\mathcal{L}_{PB} = \sum_{m=1}^M \lambda_m \rho_m \mathcal{L}_{\mathcal{G}_m} \quad (15)$$

where $\rho_m = \frac{|\mathcal{V}_m|}{|\mathcal{V}|}$ denotes the node proportion of the m -th subgraph, and λ_m serves as the adaptive balancing weight. Intuitively, this mechanism counteracts subgraph dominance by dynamically up-weighting the contribution of subgraphs with higher training losses, effectively penalizing the model for neglecting harder-to-classify structural groups. Thereby, it compels the model to treat majority and minority structural patterns equitably rather than simply fitting the dominant trend.

Datasets	Node	Edges	Features	Classes	$\mathcal{H}_G$
Cornell	183	277	1703	5	0.30
Texas	183	279	1703	5	0.06
Wisconsin	251	295	1703	5	0.18
Chameleon	2277	31371	2325	5	0.23
Squirrel	5201	198353	2089	5	0.22
Photo	7650	119081	745	8	0.83
Computers	13725	245861	767	10	0.78
Cora	2708	5278	1433	7	0.81
Pubmed	19717	44324	500	3	0.80
Citeseer	3327	4552	3703	6	0.74

Table 1: Statistics of the datasets in the experimental framework of this work.

4 Experiments

Datasets We conduct a comprehensive evaluation of CSS-GCN across ten real-world benchmark datasets, as shown in Table 1, which encompass both homophilic and heterophilic graph structures. In the case of homophilic datasets, we utilize three citation graphs specifically, namely Cora [24], Citeseer [17], and Pubmed [29], in addition to two Amazon co-purchase graphs, namely Computers and Photo. When it comes to heterophilic datasets, we adopt two Wikipedia graphs, Chameleon and Squirrel [28], as well as three webpage graphs from the WebKB collection, specifically Texas, Cornell, and Wisconsin.

Baselines We conduct a comparative analysis between the proposed CSS-GCN and 12 representative baseline methods, categorized as follows: (1) MLP [25], a feedforward neural network relying solely on node attributes; (2) Classic GNN models, including GCN [17] that captures local graph structures via convolution-like neighbor aggregation, SGC [35] which simplifies GCN computations, and GAT [33] that employs an attention mechanism to assign adaptive weights to neighboring node features; (3) Two categories of GNN models for heterophily: spectral-based methods such as GPR-GNN [7], and BernNet [12]; spatial-based methods including MixHop [1], H₂GCN [46], GloGNN [19], HopGNN [5], U-GCN [15], and NSGCN [2].

Settings All models are evaluated ten times under identical configurations of epochs, random seeds, and early stopping criteria. The hidden layer dimension is set to either 64 or 128. The hyperparameter search space encompasses learning rates selected from $\{0.05, 0.08, 0.1, 0.2, 0.3, 0.4, 0.5\}$, dropout rates ranging from 0.2 to 0.8 with an increment of 0.1, k values in $\{1, 2, 3, 4, 5\}$, weight decay values of $\{5e-4, 5e-5, 5e-6, 5e-7\}$, loss weight values in $\{0.01, 0.05, 0.1, 0.2, \dots, 1.0\}$, and the number of graph partitions in $\{1, 2, 3, \dots, 10\}$. Beyond the aforementioned settings, for baseline models, we first adopt their default hyperparameters as specified in their original papers, followed by meticulous fine-tuning of these parameters to attain superior performance. Specifically, all models are implemented using the PyTorch and PyTorch Geometric deep learning libraries, with the Adam optimizer employed for training. All experiments are conducted on machines equipped with a 4070 GPU and a 13600K CPU.

Node Classification We evaluate the node classification performance of CSS-GCN in comparison with 12 representative baseline methods across 10 benchmark datasets, with the average classification results presented in Table 2. Several key observations can be drawn from these results, and as indicated in Table 2, CSS-GCN achieves superior performance over most of the baseline methods. On homophilic datasets such as Cora and Pubmed, CSS-GCN delivers competitive accuracy despite strong performances from specialized models like GPR-GNN and HopGNN. For Citeseer, Photo, and Computers, datasets exhibiting moderate-to-high homophily, CSS-GCN consistently attains state-of-the-art results, surpassing even advanced baselines. This underscores its robustness in leveraging both structural and feature-based information within homophilic environments.

	Chameleon	Cora	Citeseer	Wisconsin	Texas	Cornell	Photo	Computers	Pubmed	Squirrel
MLP	46.56	77.29	76.56	92.50	91.97	91.80	89.44	80.66	85.65	31.24
GCN	63.98	87.05	78.48	71.38	77.84	70.08	92.21	87.04	87.14	47.36
SGC	60.09	87.28	79.85	66.63	73.77	65.57	93.06	86.50	86.45	45.36
GAT	63.69	88.44	80.63	67.50	80.00	78.67	94.36	89.03	86.65	47.83
GPR-GNN	65.65	<u>88.87</u>	80.00	92.88	92.95	89.51	93.62	88.95	89.50	53.04
BernNet	67.44	88.42	79.77	94.63	93.11	<u>92.25</u>	93.56	87.57	88.89	50.64
MixHop	50.37	87.95	79.97	70.88	90.33	78.69	91.38	86.89	86.46	36.40
U-GCN	59.43	88.24	80.01	89.50	86.72	85.25	94.16	88.83	88.97	36.63
H2GCN	58.01	88.41	79.30	92.25	89.51	89.34	93.56	88.83	89.56	34.28
GloGNN	67.00	88.41	80.41	94.00	90.00	90.98	94.13	89.83	<u>89.58</u>	57.85
HopGNN	66.85	88.93	80.15	88.88	90.33	89.67	95.34	88.91	90.02	47.89
NSGCN	<u>74.00</u>	87.96	<u>81.26</u>	<u>95.63</u>	<u>93.44</u>	91.48	<u>95.50</u>	<u>90.29</u>	88.99	<u>69.27</u>
CSS-GCN	74.42	88.19	81.80	96.38	94.43	94.10	95.74	90.83	89.05	69.79

Table 2: Results on real-world benchmark datasets: Mean accuracy (%). Boldface and underline numbers mark the top-1 and top-2 results, respectively.

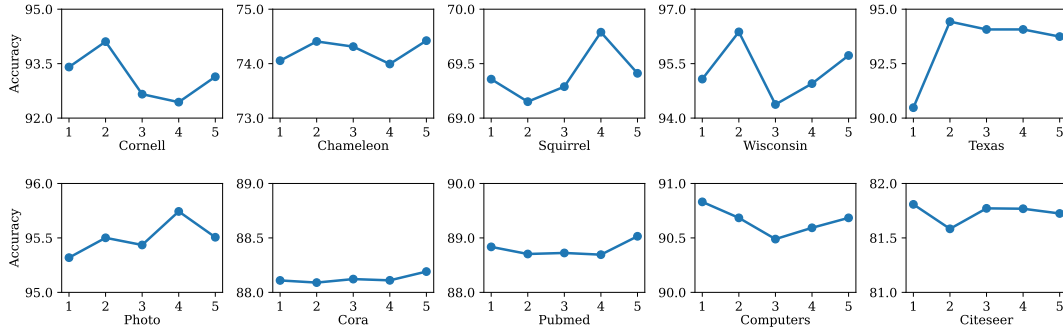


Figure 2: Analysis of Partition Number.

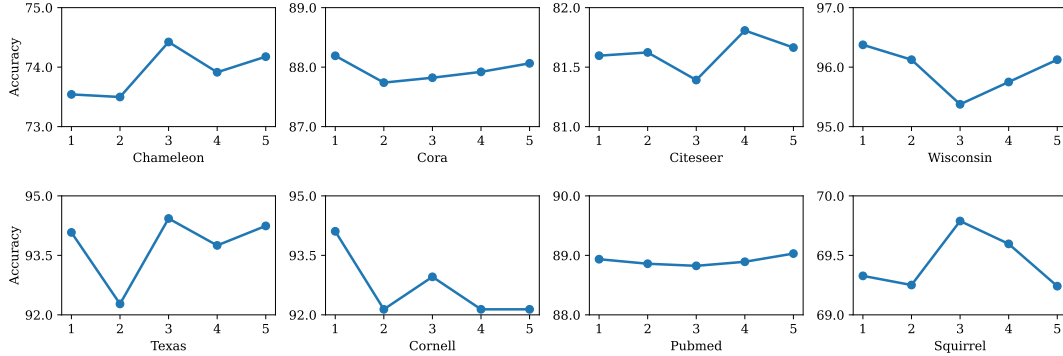


Figure 3: Analysis of Ego-Network Hop.

Remarkably, CSS-GCN demonstrates strong performance on heterophilic datasets where conventional GNNs often struggle. In Wisconsin, Texas, and Cornell, characterized by high-quality node features but low neighborhood homophily, it consistently outperforms all baselines, including models specifically designed for heterophily. For the challenging Chameleon and Squirrel datasets, where limited homophily and noisy features impede most methods, CSS-GCN achieves substantial gains. Its ability to extract critical signals from

complex heterophilic structures is evidenced by substantial margins over the closest competitors.

Across diverse graph properties, CSS-GCN demonstrates unparalleled versatility. It achieves top-tier results on seven datasets, with particularly dominant performance on heterophilic graphs. Even on datasets where it does not rank first, such as Pubmed, it remains highly competitive. These outcomes validate the efficacy

of CSS-GCN in adaptively synthesizing structural and feature information, establishing it as a highly versatile framework for node classification.

Analysis of Ego-Network Hop In the process of constructing ego-networks, the selection of an optimal number of hops plays a pivotal role in determining the final model performance. As visually depicted in Figure 3, the accuracy trajectories of different datasets fluctuate divergently as the hop count increments from 1 to 5. Specifically, Chameleon’s accuracy exhibits minor fluctuations with varying hop counts but fundamentally remains at a consistent level overall; in comparison, Cora performs with remarkable stability, showing negligible difference in accuracy across different hop counts. In contrast, Wisconsin achieves optimal performance at hop count 1 and subsequently fluctuates to a noticeable extent; meanwhile, Squirrel’s accuracy, though exhibiting variations, persists at a relatively low level. Conversely, Texas and Cornell show significantly more obvious performance changes under different hop counts, evidencing a pattern of peaking at a certain hop count while declining at others. In general, the choice of hop count profoundly influences the model’s ability to effectively capture node neighbor information, thereby affecting the final accuracy, and this impact is intrinsically correlated with dataset characteristics, indicating that there is no universally optimal hop count applicable to all datasets.

Analysis of Partition Number As illustrated in Figure 2, for datasets characterized by heterogeneous graphs, the performance exhibits minor fluctuations across the evaluated partition numbers, with certain datasets demonstrating slight improvements or variations while simultaneously maintaining relatively high overall stability. In contrast, the homogeneous graphs display consistently robust performance across all partition numbers, indicating a lack of sensitivity to changes in the partitioning strategy. This observation suggests that while partition number adjustments may introduce subtle variations in heterogeneous settings, the robustness of the models remains largely unaffected in homogeneous environments.

Analysis of Different Representations Table 3 provides a comprehensive summary of the mean accuracy results of different methods on node classification tasks, where *w/o-p* denotes the setting with potential neighbor information removed, and *w-p*, *w-f*, *w-s* correspond to the experimental scenarios using exclusively potential neighbors, only first-order neighbors, and only self-representations, respectively. The *w/o-p* setting, which removes potential neighbor information, consistently outperforms *w-p* on most datasets, indicating that exclusive reliance on potential neighbor information may not effectively improve model performance, and the strategic and rational use of other information sources may be more critical. The *w-f* and *w-s* configurations show commendable performance on some datasets, but overall they are still inferior to the full CSS-GCN and *w/o-p*, further demonstrating the importance of synergistically and comprehensively utilizing multiple information sources to enhance node classification performance.

Ablation Study Table 4 presents the results of the ablation study, where *w/o* loss denotes the model variant without using Partition Balanced Loss, *w/o* parts corresponds to the version without graph partition, and *w/o* ego refers to the setting without constructing ego-networks. It can be observed that the full CSS-GCN model achieves the best performance across all datasets, with accuracy

Datasets	Chameleon	Cora	Citeseer	Squirrel
<i>w/o-p</i>	<u>71.66 ± 1.44</u>	86.84 ± 0.61	<u>79.74 ± 0.60</u>	<u>66.11 ± 0.69</u>
<i>w-p</i>	45.27 ± 1.47	75.62 ± 1.18	76.35 ± 1.98	32.57 ± 0.83
<i>w-f</i>	66.15 ± 1.55	<u>87.28 ± 0.71</u>	65.45 ± 4.16	62.85 ± 0.75
<i>w-s</i>	64.31 ± 1.09	77.81 ± 0.57	50.53 ± 2.80	62.85 ± 0.71
CSS-GCN	74.42 ± 1.31	88.19 ± 0.62	81.80 ± 0.44	69.79 ± 0.87

Table 3: Result of Different Representations

	Chameleon	Cora	Citeseer	Wisconsin	Cornell	Squirrel
<i>w/o</i> attention	73.48	87.61	81.10	95.25	91.63	68.81
<i>w/o</i> parts	74.00	87.91	81.40	94.38	92.95	69.29
<i>w/o</i> ego	73.98	87.56	81.29	92.38	93.61	69.24
CSS-GCN	74.42	88.19	81.80	96.38	94.10	69.79

Table 4: Ablation study results.

significantly higher than that of each ablation variant. This indicates that the three key components of the model, namely Partition Balanced Loss, graph partition, and ego-networks construction, all contribute to overall performance. Removing any one of these components leads to a performance drop to varying degrees. Although the impact of each component varies across datasets, the synergistic effect of the three components is generally an important reason for excellent performance of CSS-GCN.

5 Conclusion

This paper proposes a novel Contextual Structure Enhanced Selective Graph Convolutional Network that effectively models and unifies representations for both homophilic and heterophilic networks for node classification. Specifically, we introduce a Contextual Structure Enhanced (CSE) approach that precisely identifies potentially similar neighbors through a comprehensive node similarity measure capturing intrinsic structural and attribute relationships, while simultaneously constructing k-hop ego-networks to capture fine-grained local contextual structures. Crucially, the CSE module emphasizes connections to contextually similar neighbors based on feature-topology affinity, thereby mitigating aggregation noise for heterophilic nodes while preserving essential intra-class information for homophilic nodes. Building upon this foundation, we design a sophisticated multi-path fusion mechanism that dynamically aggregates three key information sources—potential neighbors, directly connected first-hop neighbors, and ego-representations—via multi-head attention scheme. By learning and capturing node-specific attention distributions, our framework enables precise modeling of complex homophily and heterophily patterns. Extensive experiments conducted on 10 real-world datasets with diverse attributes and varying homophily levels demonstrate the superiority of CSS-GCN. The synergistic integration of graph partitioning and adaptive multi-path fusion significantly enhances robustness against structural shifts, validating consistent performance gains in complex network environments.

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